Conditional Probability

Lecture 3

Topics in this lecture

- Conditional probability
- Independence of events
- Sequence of independent experiments

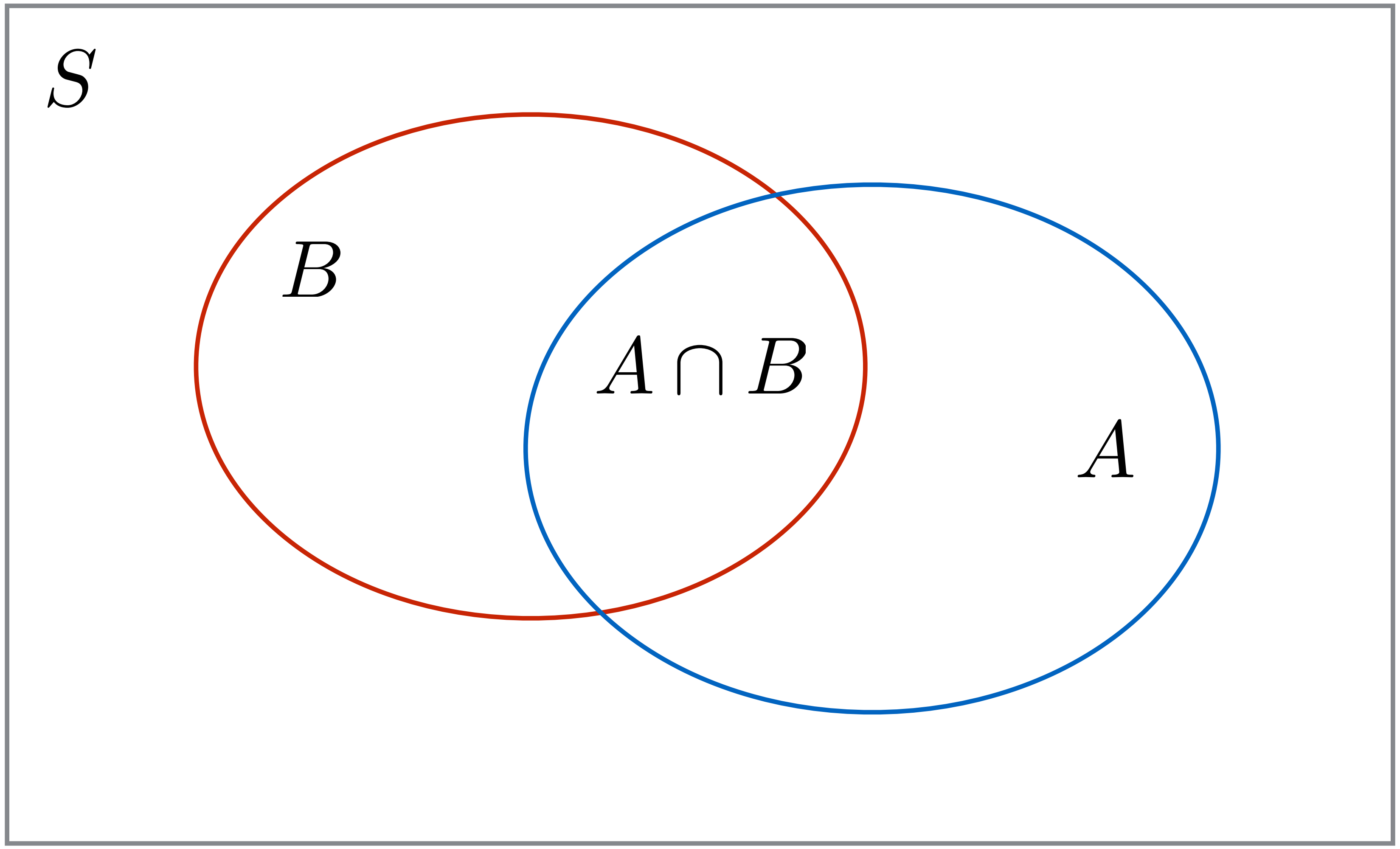
Conditional Probability

- Does knowledge about the occurrence of one event, say B, alter the likelihood of occurrence of the other, say A?

- Conditional probability is defined as

$$\boxed{P(A|B) = \frac{P(A \cap B)}{P(B)} \quad , \quad \text{for} \quad P(B) > 0}$$

Graphical Interpretation



$$\frac{n_{A\cap B}}{n_B} = \frac{n_{A\cap B}/n}{n_B/n} \rightarrow \quad \frac{P(A\cap B)}{P(B)}$$

Alternative Expressions

$$P(A \cap B) = P(A|B)P(B)$$

$$P(A \cap B) = P(B|A)P(A)$$

Example 3.1

- Toss a coin 3 times. What is the conditional probability of getting all heads given that

  (a) The first toss lands on head

  (b) At least 1 toss lands on head

Example 3.2

- Roll a die twice and observe the sum of the marks from two rolls. Find the conditional probability that the sum of the marks is greater than 7 given that $\Omega = \{2, 3, \ldots, 12\}$ | $A = \text{sum} > 7 = \{8, 9, 10, 11, 12\}$

  $\Omega = \begin{Bmatrix} (1,1), (1,2), \ldots, (1,6) \\ (2,1), (2,2), \ldots, (2,6) \\ \vdots \\ (6,1), (6,2), \ldots, (6,6) \end{Bmatrix}$

  (a) The outcome of the first roll is 3 $B = \{(3,1), (3,2), (3,3), \ldots (3,6)\}$

$A = \text{sum} > 7 = \begin{Bmatrix} (2,6), (3,5), (4,4) \\ (6,2), (5,3) \; 8 \\ 9 \; (3,6), (4,5), (5,4), (6,3) \\ 10 \; (4,6), (5,5), (6,4) \\ 11 \; (5,6), (6,5) \\ 12 \; (6,6) \end{Bmatrix}$

$P(A) = \frac{15}{36} = \frac{5}{12}$

$P(A|B) = \frac{P(A \cap B)}{P(B)}$

$P(B) = \frac{6}{36}$

$A \cap B = \{(3,5), (3,6)\}$

$P(A \cap B) = \frac{2}{36}$

Example 3.2

- Roll a die twice and observe the sum of the marks from two rolls. Find the conditional probability that the sum of the marks is greater than 7 given that $\Omega = \{2, 3, \ldots, 12\}$ | $A = \text{sum} > 7 = \{8, 9, 10, 11, 12\}$

$$\Omega = \begin{Bmatrix} (1,1), (1,2), \ldots, (1,6) \\ (2,1), (2,2), \ldots, (2,6) \\ \vdots \\ (6,1), (6,2), \ldots, (6,6) \end{Bmatrix}$$

$$A = \text{sum} > 7 = \begin{Bmatrix} 8 & (2,6), (3,5), (4,4), (6,2), (5,3) \\ 9 & (3,6), (4,5), (5,4), (6,3) \\ 10 & (4,6), (5,5), (6,4) \\ 11 & (5,6), (6,5) \\ 12 & (6,6) \end{Bmatrix}$$

$$P(A) = \frac{15}{36} = \frac{5}{12}$$

  (b) The outcome of the first roll is 6

Example 3.3

- A ball is selected from an urn containing two black balls, numbered 1 and 2, and two white balls, numbered 3 and 4. The number and color of the ball is noted. Find $P(A|B)$ and $P(A|C)$ where A, B, C are
  - A = "Black ball is selected"
  - B = "Even-numbered ball is selected"
  - C = "Number of ball is greater than 2"

Total Probability Theorem

- Let B_1, B_2, …, B_n be *mutually exclusive* events whose union equals the sample space S. We refer to these sets as a partition of S

- The event A can be represented as

$$\begin{aligned} A &= A \cap S = A \cap (B_1 \cup B_2 \cup \cdots \cup B_n) \\ &= (A \cap B_1) \cup (A \cap B_2) \cup \cdots \cup (A \cap B_n) \end{aligned}$$

- The probability of event A can be written as

$$P(A) = P(A \cap B_1) + P(A \cap B_2) + \cdots + P(A \cap B_n)$$

$$P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \cdots + P(A|B_n)P(B_n)$$

Example 3.4

- An insurance company believes that people can be divided into two classes: those who are accident prone and those who are not. The company's statistics show that an accident-prone person will have an accident at some time within a fixed 1-year period with probability 0.4, whereas this probability decreases to 0.2 for a person who is not accident prone. If we assume that 30 percent of the population is accident prone, what is the probability that a new policyholder will have an accident within a year of purchasing a policy?

Bayes' Rule

- Let B_1, B_2, …, B_n be a partition of a sample space S. Suppose that event A occurs, the probability of event B_j given A is

$$P(B_j|A) = \frac{P(A \cap B_j)}{P(A)} = \frac{P(A|B_j)P(B_j)}{\sum_{k=1}^{n} P(A|B_k)P(B_k)}$$

- P(B_j)'s are called "a priori" probabilities

- P(B_j | A) are called "a posteriori" probabilities

Example 3.5

- From Example 3.4, suppose that a new policyholder has an accident within a year of purchasing a policy. What is the probability that he or she is accident prone?

Example 3.6

- A laboratory blood test is 95 percent effective in detecting a certain disease when it is, in fact, present. However, the test also yields a "false positive" result for 1 percent of the healthy persons tested. (That is, if a healthy person is tested, then, with probability 0.01, the test result will imply that he or she has the disease.) If 0.5 percent of the population actually has the disease, what is the probability that a person has the disease given that the test result is positive?

A = Positive

A' = negative

B = disease

B' = No disease

$P(A|B) = 0.95$

$P(A|B') = 0.01$

$P(B) = 0.005$

$P(B') = 0.995$

$$P(B|A) = \frac{P(A|B)P(B)}{P(A|B)P(B) + P(A|B')P(B')}$$

$$= \frac{0.95 \cdot 0.005}{(0.95 \cdot 0.005) + (0.01 \cdot 0.995)}$$

Independence of Events

- Events *A* and *B* are independent if the knowledge of the occurrence of one event does not alter the likelihood of the other.

- Formally put

$$P(A) = P(A|B) = \frac{P(A \cap B)}{P(B)}$$

- Thus, events *A* and *B* are *independent* if

$$\boxed{P(A \cap B) = P(A)P(B)}$$

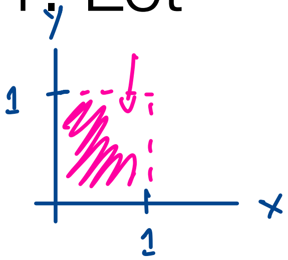
Example 3.7

- A ball is selected from an urn containing two black balls, numbered 1 and 2, and two white balls, numbered 3 and 4. Let the events A, B, and, C be defined as
  - A = "Black ball is selected"
  - B = "Even-numbered ball is selected"
  - C = "number of ball is greater than 2"
- Are events A and B independent?
- Are events A and C independent?

Example 3.8

- Two numbers x and y are selected at random between 0 and 1. Let events A, B, and C be defined as follows

$\Omega = \{(x,y) : 0 < x < 1, 0 < y < 1\}$



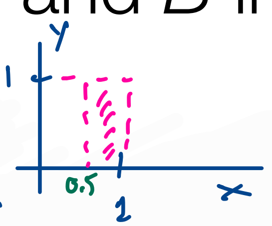
$$A = \{x > 0.5\}$$

$$B = \{y > 0.5\}$$

$$C = \{x > y\}$$

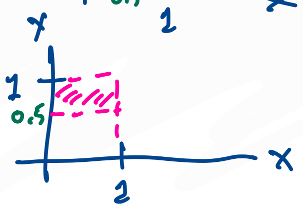
- Are events A and B independent? $P(A \cap B) = P(A)P(B)$

$A = \{x > 0.5\} =$



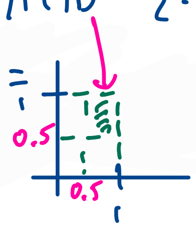
$P(A) = \frac{0.5}{\underset{\Omega}{1}} = \frac{1}{2}$

$B = \{y > 0.5\} =$



$P(B) = \frac{0.5}{\underset{\Omega}{1}} = \frac{1}{2}$

$A \cap B = \{x > 0.5, y > 0.5\} =$



$P(A \cap B) = \frac{1}{4}$

$P(A \cap B) = \frac{1}{4}$

$P(A)P(B) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$

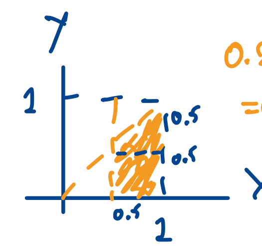
Example 3.8

- Two numbers x and y are selected at random between 0 and 1. Let events A, B, and C be defined as follows

$$A = \{x > 0.5\}$$

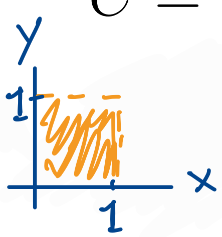
$$B = \{y > 0.5\}$$

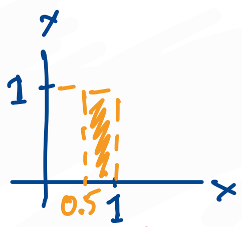
$$C = \{x > y\}$$

$A \cap C = \{x > 0.5,\ x > y\} =$ 

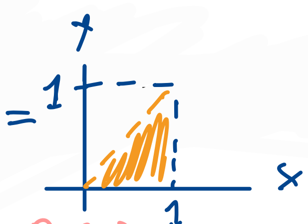
$0.25 + 0.125$
$= 0.375$

$P(A \cap C) = \frac{3}{8}$

$\Omega =$ 

$A =$ 

$P(A) = 0.5 = \frac{1}{2}$

$C =$ 

$P(C) = 0.5 = \frac{1}{2}$

- Are events A and C independent? $P(A \cap C) = P(A)P(C)$

$P(A \cap C) = \frac{3}{8}$

$P(A)P(C) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$

Sequence of Independent Experiments

- Suppose that a random experiment consists of performing experiments E_1, E_2, …, E_n. Let A_1, A_2, …, A_n be events such that A_k concerns only the outcome of the k-th sub-experiment. If the sub-experiments are independent, then it is reasonable to assume that events A_1, …, A_n are independent.

- It follows that

$$P(A_1 \cap A_2 \cap \cdots \cap A_n) = P(A_1)P(A_2)\ldots P(A_n)$$

Example 3.9

- Box 1 contains 5 red balls and 5 yellow balls. Box 2 contains 3 red balls and 7 white balls. Select a ball from each box. Find the probability that the 2 red balls are selected.

Example 3.10

- Suppose that 10 numbers are selected at random from the interval [0,1]. Find the probability that the first 5 numbers are less than 1/4 and the last 5 numbers are greater than 1/2.